Interpreting GPT: The Logit Lens

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nostalgebraist

Why a LessWrong post?

- Verbalizable Representations Form a Global Workspace in Language Models (2026) cited it, and I was struck by how familiar the poster's name was.
- Turns out this is the guy who wrote the AO3 fic The Northern Caves (2015), which is why I recognized his name.
- Turns out this post is extremely widely cited in AI mechanistic interpretability.
- In particular, it is cited by Eliciting Latent Predictions from Transformers with the Tuned Lens (2023), which builds on it to create the tuned lens.
- Verbalizable Representations Form a Global Workspace in Language Models then builds on that to create the Jacobian lens.

Background on GPT Structure

- GPT takes in and puts out tokens, numerical representations of text. There are 50,257 tokens in the vocabulary, and the output is a probability distribution across all possible tokens.
- Between input and output, tokens are represented as embedding vectors, vectors of size 1600 in GPT-2 1558M.
- There is an “embedding matrix” W which converts tokens to embeddings and vice versa.
- The GPT model is a series of 48 transformers back to back.
- Each transformer takes in an input token i and outputs a size 1600 vector.
- That vector can be expanded with the embedding matrix to produce a size 50,257 vector, where each number is a logit, a score indicating the confidence of the model that token i+1 will be the token with the corresponding index. These aren’t probabilities: they can be outside the range [0, 1]. To convert them to probabilities you apply softmax.

Overview

- GPT probs are the result of softmax applied to logits of the final transformer, but you can also look at the logits of non-final transformer layers.
- One might presume that there's smooth change from input to output across the layers. That is, top-k logits for some k include the input token in early layers and include the output token in late layers, with the change happening gradually in the middle layers.
- In reality, the first layer (input layer) yields top-1s basically identical to the inputs, while the first hidden layer (e.g. the one after) seems completely disconnected from the inputs: much closer to the outputs than inputs, but not particularly close to either.
- Early output seems nonsensical, middle output clearly vaguely related to final output, late output converges rapidly on final output, though for some input tokens this trend does not hold.

Overview

The fact that only the first layer reflects the input tokens indicates that inputs are immediately converted to an intermediate representation.

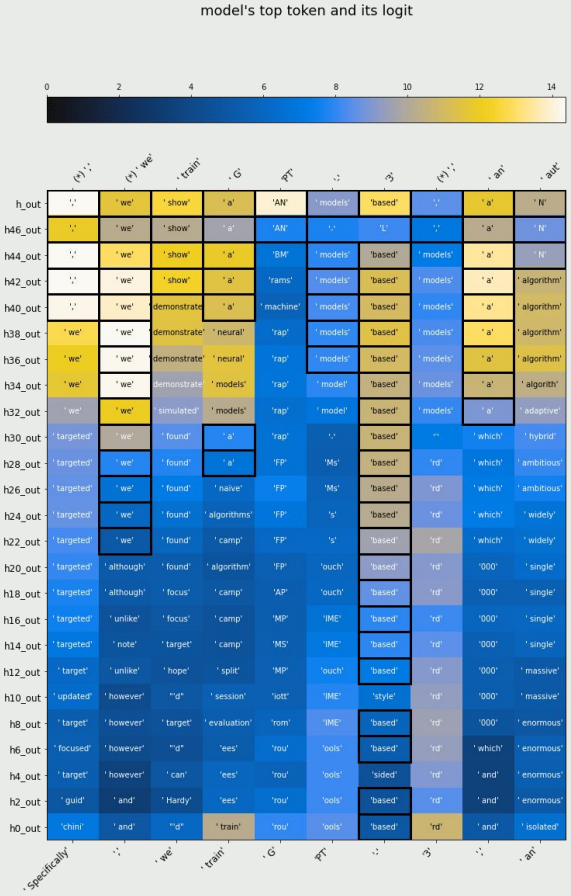
> *More speculatively, this suggests that GPT mostly 'thinks in predictive space,' immediately converting inputs to predicted outputs, then refining guesses in light of other guesses that are themselves being refined.*

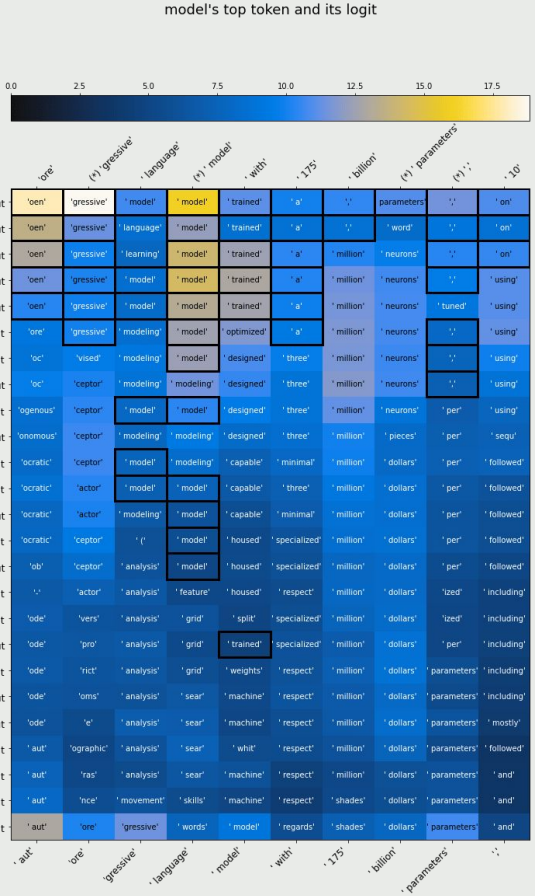
The Logit Lens

- “The logit lens” simply refers to looking at the outputs of each transformer layer, rather than the final layer.
- Use the embedding matrix to turn the output into the same form as the final logits.
- This lets us see how the model is processing inputs across the whole depth.

Logits

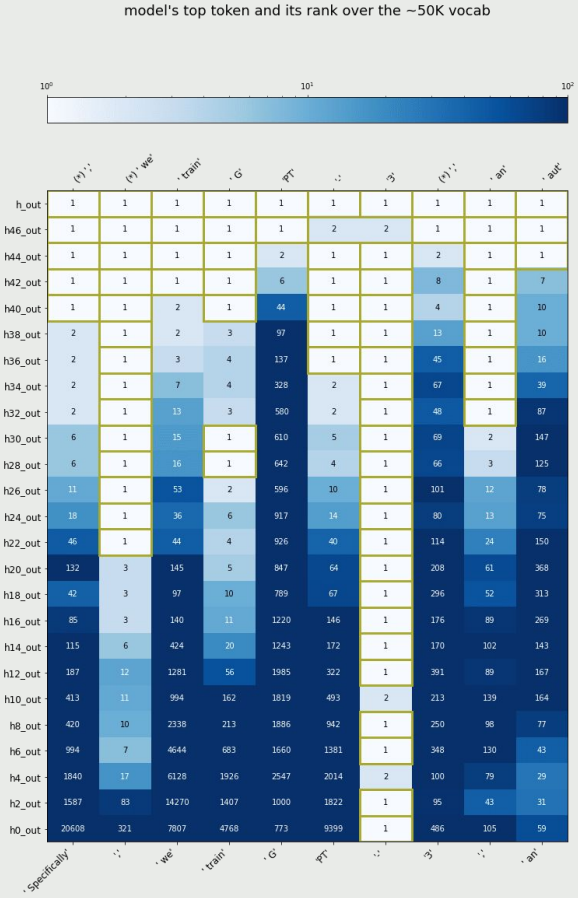
- Final predictions mostly don't get locked in until the final few layers.
- Early predictions are quite different from later guesses but generally makes sense as tokens which could follow the input token (but maybe not in this context).
- Early layers' top tokens have smaller logits than later ones, even if the token is the same (see 'based' and 'model').

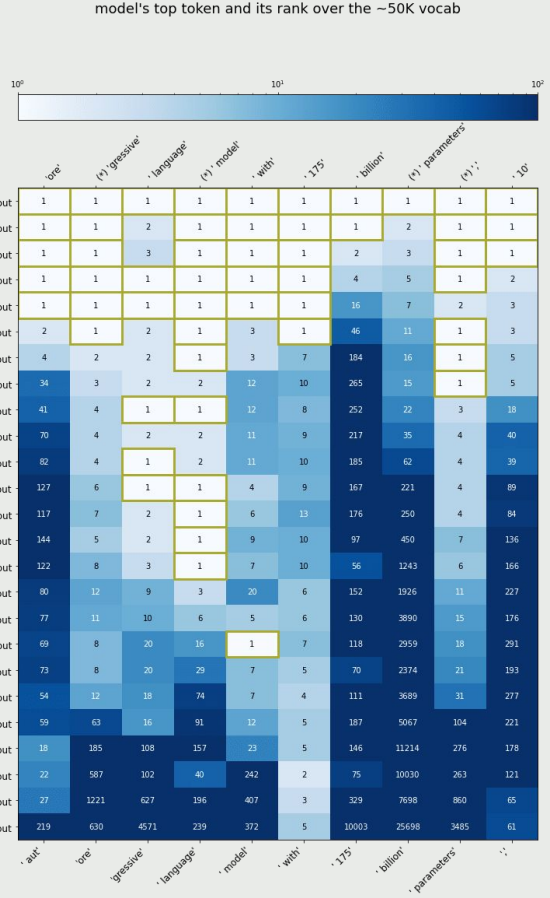




Ranks

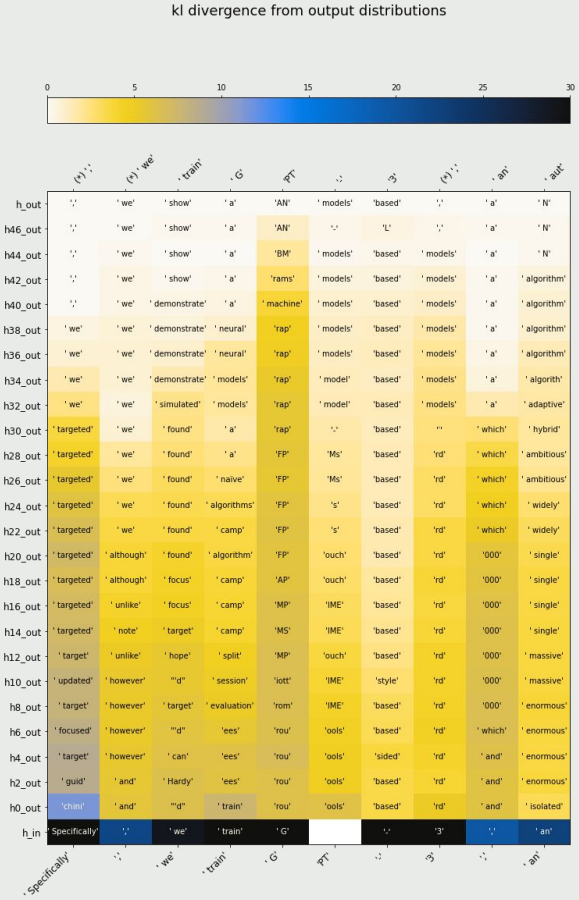
- What if we look at where the final prediction is ranked at each stage?
- The final prediction is generally near the top rank by the middle layers, even if it only gets to #1 at the very end.

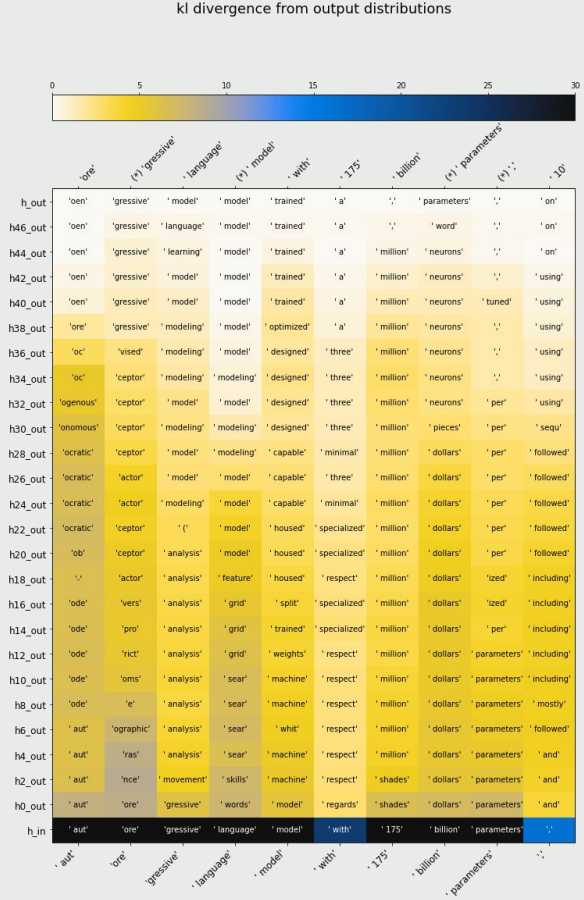




KL divergence

- KL divergence is a measure of how dissimilar two probability distributions are. It's a non-negative number, and the higher the number, the more dissimilar the two are.
- Previous charts didn't show the input layer "because they're so different from everything else, they distract the eye!"
- These charts compare the output distributions to the distribution at each layer



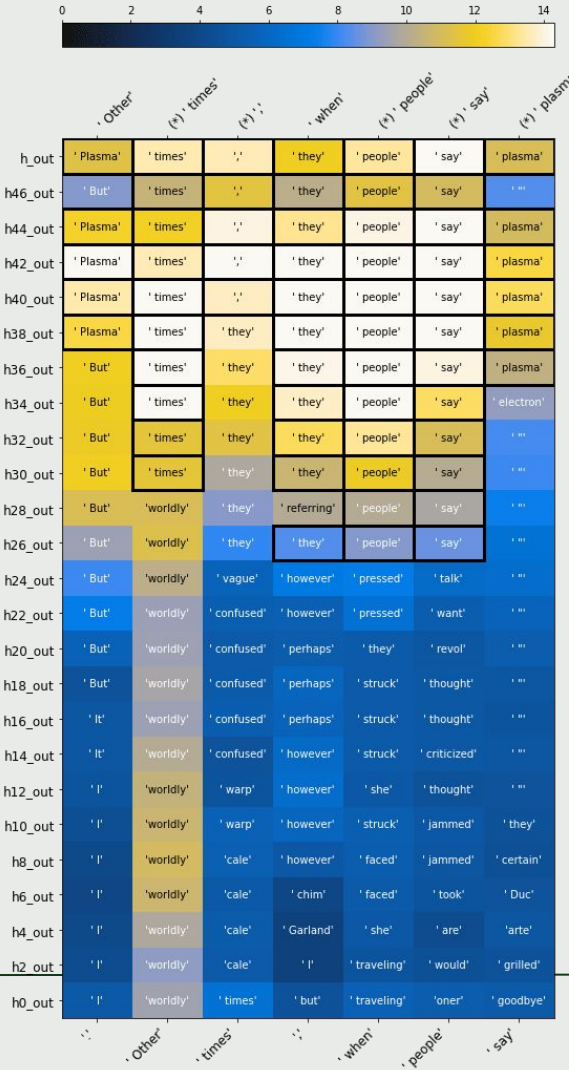


Copying a rare token

- What happens if a token is a) repeated and b) generally improbable?

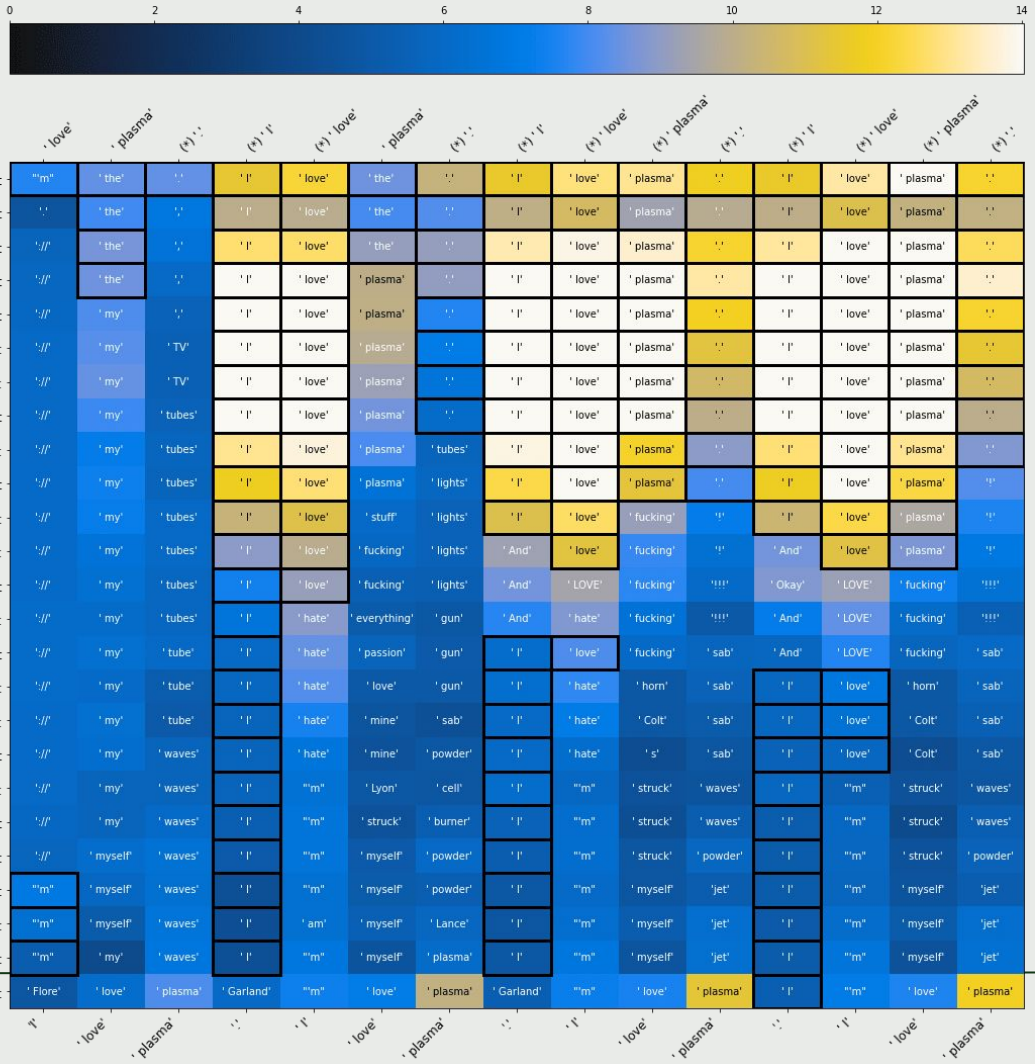
  “Sometimes, when people say plasma, they mean a state of matter. Other times, when people say ____”

- Plasma is a rare word, so in most contexts it should have a low probability. In context, it’s clear to a human that the next word should be “plasma”, because of the repetitive pattern.



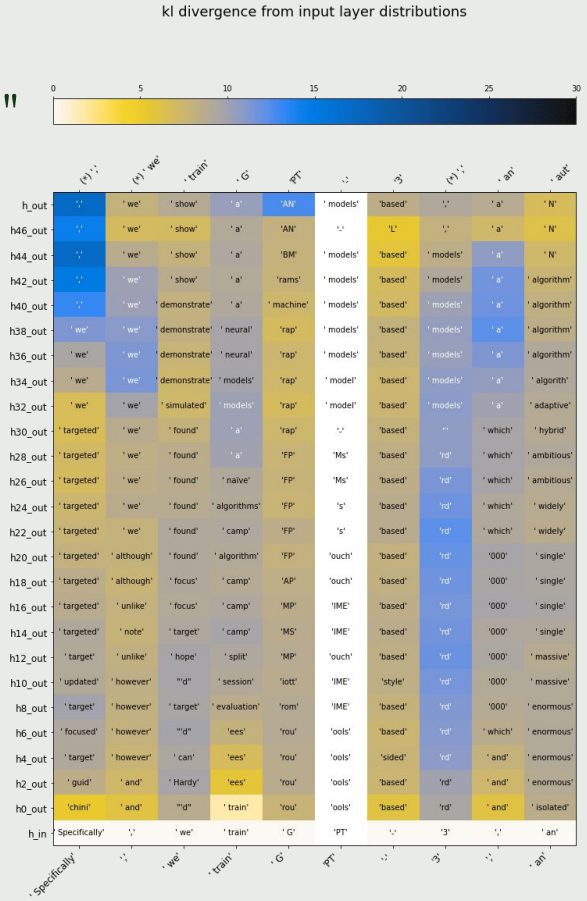
Extreme repetition

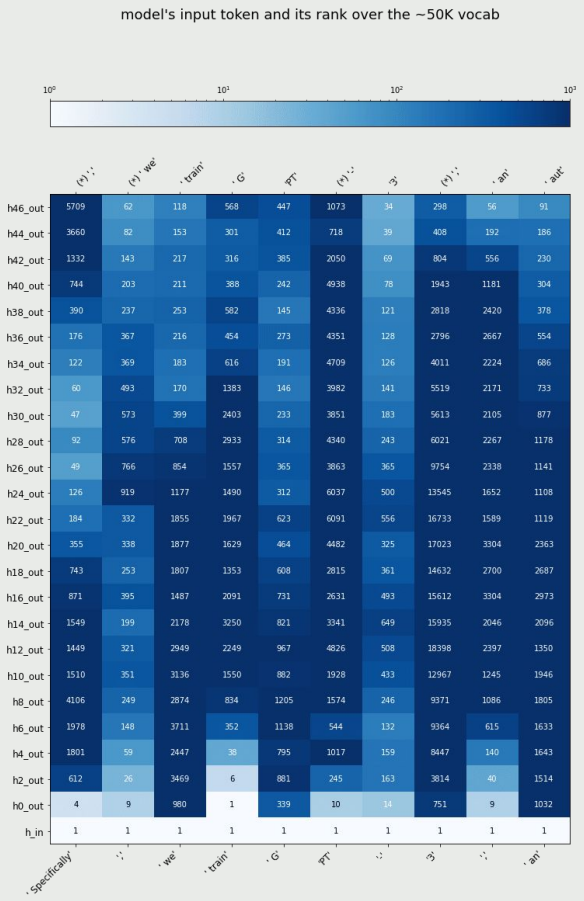
Upper half (later layers) notices repetition, but lower half doesn't seem to



Addendum:
More on "input discarding"

- In some comments, it was pointed out that maybe taking the KL with respect to the output is misleading - look at input instead.
- Input probability and rank tends to drop off sharply (but not as sharply as before) and then gradually resurface towards the end.
- I don't know why that column is all white.

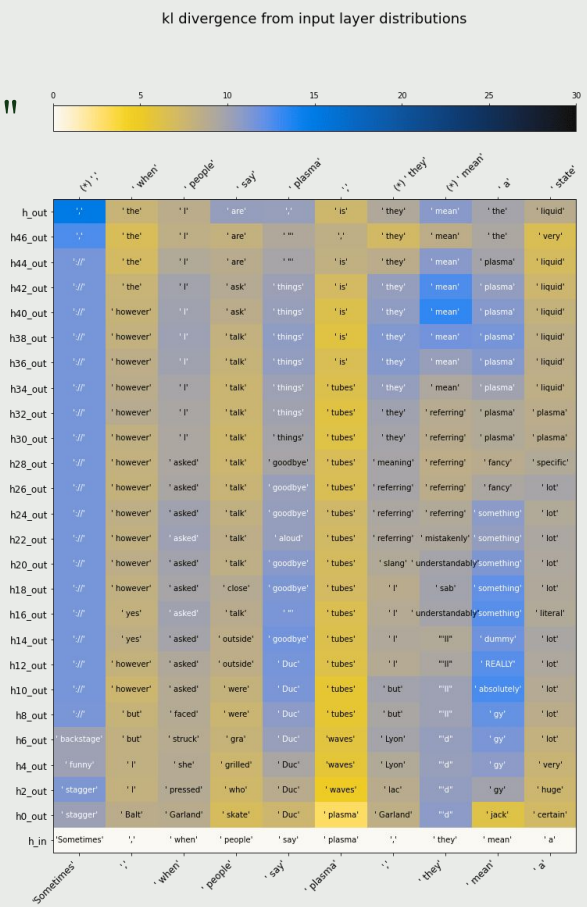


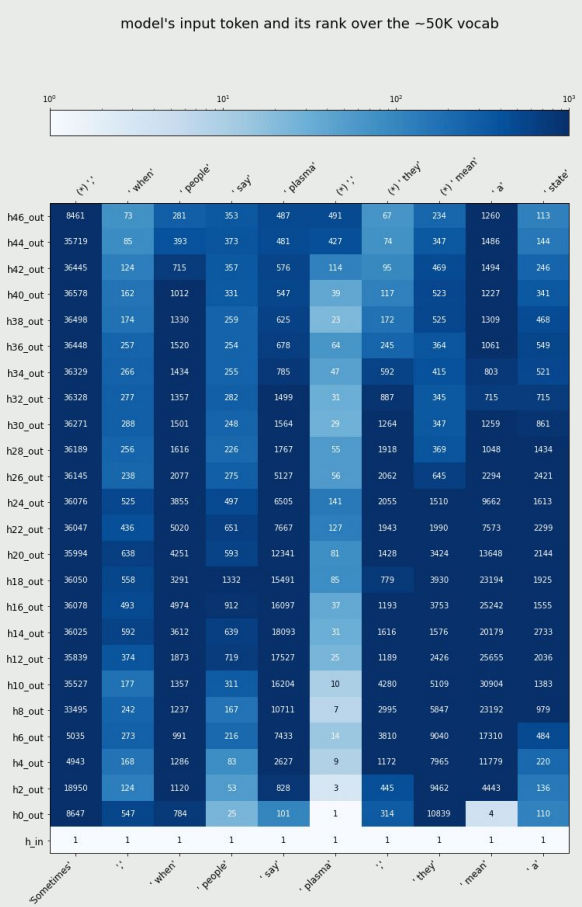


Addendum:
More on "input discarding"

Plasma is pretty well preserved the whole way through.

"My intuitive guess is that the rarity, or (in some sense) 'surprisingness,' of the token causes early layers to preserve it: this would provide a mechanism for providing raw access to rare tokens in the later layers, which otherwise only be looking at more plausible tokens that GPT had guessed for the corresponding positions."





Why? Is this surprising?

“I would have expected a U-shaped pattern, where the early layers mostly look like the input, the late layers mostly look like the output, and there's a gradual ‘flip’ in the middle between the two perspectives. Instead, the input space immediately vanishes, and we're in output space the whole way.”

…

“Or, maybe there's some sort of ‘hidden’ invertible relationship between

- *the embedding of a given token, and*
- *the model's prior for what token comes after it (given no other information)*

so that a token like ‘plasma’ is kept around from the input -- but not in the form ‘the output is plasma,’ instead in the form ‘the output is [the kind of word that comes after plasma].’”

Legacy

- People don't seem to really cite nostalgebraist's analysis that much, but the logit lens as a tool has been very influential.
- Eliciting Latent Predictions from Transformers with the Tuned Lens (2023)
  - *"We build on the "logit lens" (nostalgebraist, 2020), an early exiting technique that directly decodes hidden states into vocabulary space using the model's pretrained unembedding matrix. We find the logit lens to be unreliable (Section 2), failing to elicit plausible predictions for models like BLOOM (Scao et al., 2022) and GPT Neo (Black et al., 2021). Even when the logit lens appears to work, its outputs are hard to interpret due to representational drift: features may be represented differently at different layers of the network. Other early exiting procedures also exist (Schuster et al., 2022), but require modifying the training process, and so can't be used to analyze pretrained models."*
- Verbalizable Representations Form a Global Workspace in Language Models (2025)
  - *"Our results make use of a new interpretability technique called the Jacobian lens (J-lens) … The J-lens can be understood as a principled refinement of the logit lens [28]. While the logit lens assumes that representations use the same coordinates in all layers, the Jacobian lens corrects for representational changes that take place across layers, allowing it to uncover meaningful information in earlier layers where the logit lens produces uninterpretable readouts."*